

OOPS: Task Management System

1. The task management system should allow users to create, update, and delete tasks.
2. Each task should have a title, description, due date, priority, and status (e.g., pending, in progress, completed).
3. Users should be able to assign tasks to other users and set reminders for tasks.
4. The system should support searching and filtering tasks based on various criteria (e.g., priority, due date, assigned user).
5. Users should be able to mark tasks as completed and view their task history.
6. The system should handle concurrent access to tasks and ensure data consistency.
7. The system should be extensible to accommodate future enhancements and new features.

Question 5: Online Course Video Tracker

Problem Statement

An online learning platform tracks video views as a space-separated string. Each word represents a video title viewed by a student.

Write a function:

```
def analyze_views(views: str) -> dict:
```

Tasks

Your function should:

- 1. Handle empty input**
 - Return {} if input is empty
- 2. Normalize input**
 - Convert all video titles to lowercase
- 3. Split into a list of viewed videos**
- 4. Create a set of unique videos**
- 5. Count views per video**

6. Calculate

- Total number of views
- Number of unique videos
- Most watched video
- Least watched video

7. Return a dictionary with keys

- "views"
- "unique_videos"
- "total_views"
- "view_counts"
- "most_watched"
- "least_watched"

Example Input

views = "Intro Python intro Java Python intro"

Example Output

```
{  
  "views": ["intro", "python", "intro", "java", "python", "intro"],  
  "unique_videos": {"intro", "python", "java"},  
  "total_views": 6,  
  "view_counts": {"intro": 3, "python": 2, "java": 1},  
  "most_watched": "intro",  
  "least_watched": "java"  
}
```